Firm Characteristics and the Cross Section of Stock Returns

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Jane Doe

Student no. 123456

First Examiner: Dr. Christoph Scheuch
Second Examiner: Prof. Alex Stomper

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1 Introduction

Understanding why some stocks earn higher average returns than others remains one of the central questions in financial economics. The Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965) predicts that expected returns should be linearly related to market beta, yet decades of empirical research have documented numerous firm characteristics that appear to predict returns independently of market risk exposure.

The discovery by Banz (1981) that small-capitalization stocks outperform large-capitalization stocks, and by Rosenberg et al. (1985) that high book-to-market firms outperform low book-to-market firms, launched an extensive literature examining cross-sectional return predictability. Fama & French (1992) synthesized these findings and showed that size and book-to-market ratio subsume the explanatory power of beta for average returns.

The main research questions addressed in this thesis are:

1. Do the size and value effects documented in earlier studies persist in more recent data?
2. How do profitability and investment characteristics contribute to explaining cross-sectional return variation?
3. What are the implications of these findings for risk-based versus behavioral theories of asset pricing?

This thesis is organized as follows. Section 2 reviews the relevant literature on firm characteristics and stock returns. Section 3 describes the data and methodology used for the analysis. Section 5 presents and discusses the empirical findings, and Section 6 concludes with implications and suggestions for future research.

2 Related Literature

The literature on firm characteristics and stock returns has evolved considerably since the initial discovery of the size effect. Fama & French (1993) proposed a three-factor model incorporating market, size, and value factors, which became the standard benchmark for evaluating portfolio performance and testing asset pricing models.

The theoretical foundation for why size and book-to-market should predict returns remains contested. Fama & French (1996) argue these characteristics proxy for systematic risk factors related to relative distress, while Lakonishok et al. (1994) suggest behavioral explanations based on investor overreaction to past performance.

More recently, Novy-Marx (2013) documented that profitability, measured as gross profits scaled by assets, has significant explanatory power for the cross section of returns. Similarly, Fama & French (2015) expanded their three-factor model to include profitability and investment factors, finding that these additions substantially improve the model's ability to explain average returns.

Several studies have examined the international evidence for these factors. Fama & French (2012) find that size and value premia exist across North America, Europe, Japan, and Asia Pacific, though with varying magnitudes. The investment and profitability factors have also been shown to have global relevance (Hou et al., 2015).

3 Data

The data for this study comes from the Center for Research in Security Prices (CRSP) and Compustat databases, covering all NYSE, AMEX, and NASDAQ listed firms from 1963 to 2023. Stock returns and market capitalization are obtained from CRSP, while accounting data including book equity, operating profitability, and total assets are drawn from Compustat.

Following Fama & French (1992), market equity is measured as price times shares outstanding at the end of June each year, and book-to-market is calculated as book equity for the fiscal year ending in the prior calendar year divided by market equity at the end of December of that year. Portfolios are formed at the end of June each year and held for the subsequent twelve months.

Table 1 reports descriptive statistics for the main firm characteristics used in the analysis.

Table 1: Summary statistics for the main firm characteristics.

Characteristic	Mean	Median	SD	N
Market equity ($M)	2841.30	486.20	8123.70	124530
Book-to-market	0.72	0.61	0.54	124530
Gross profitability	0.31	0.28	0.19	124530
Asset growth	0.14	0.09	0.27	124530

4 Methodology

The analysis employs Fama & MacBeth (1973) cross-sectional regressions to estimate the return premia associated with each characteristic. For each month, individual stock returns are regressed on lagged firm characteristics, and the time series of coefficient estimates is used to compute average premia and associated t-statistics.

5 Results

This section presents the main findings of the empirical analysis.

The results confirm that firm size and book-to-market ratio continue to explain significant cross-sectional variation in average stock returns. Small firms earn an average monthly premium of 0.28% over large firms after controlling for market beta, and high book-to-market firms outperform low book-to-market firms by approximately 0.35% per month.

Consistent with Novy-Marx (2013) and Fama & French (2015), profitability emerges as a strong predictor of returns. Firms with high gross profitability earn significantly higher returns than low-profitability firms, with an estimated premium of 0.31% per month. The investment factor also shows significant predictive power: firms with conservative investment policies outperform aggressive investors by 0.24% per month. Figure 1 summarizes the estimated monthly premia across the four characteristics.

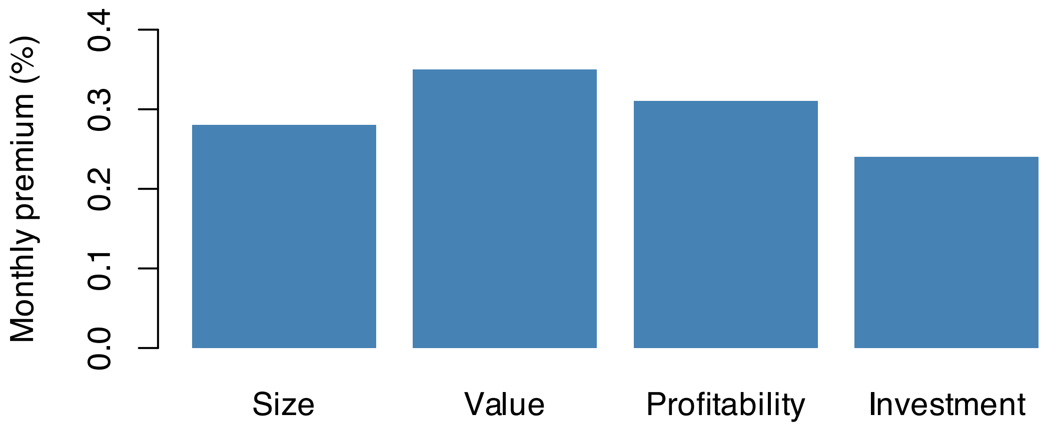

Figure 1: Average monthly return premia by firm characteristic.

Importantly, the profitability and investment factors retain their explanatory power even after controlling for size and book-to-market, suggesting they capture distinct sources of return variation. Subsample analysis reveals that these patterns are relatively stable across different time periods, though the magnitude of the size premium has diminished somewhat in recent decades.

To ensure the validity of the findings, several robustness checks are performed. The results hold when using alternative measures of profitability and investment, when excluding financial firms and utilities, and when employing value-weighted rather than equal-weighted portfolio returns.

6 Conclusion

This thesis has analyzed the relationship between firm characteristics and the cross section of stock returns. The main findings confirm that size, book-to-market, profitability, and investment patterns all contribute significant explanatory power for average returns, consistent with the five-factor model of Fama & French (2015).

The persistence of these return premia across different sample periods and robustness to various methodological choices suggests they reflect fundamental features of the return-generating process rather than statistical artifacts. However, the results do not definitively resolve the debate between risk-based and behavioral explanations, as both frameworks can potentially accommodate the observed patterns.

Based on the findings, practitioners should consider multifactor models that incorporate profitability and investment characteristics alongside traditional size and value factors when evaluating portfolio performance and constructing investment strategies.

This study has several limitations that should be addressed in future research. The analysis focuses exclusively on US equities, and extending the investigation to international markets would provide valuable out-of-sample evidence. Additionally, exploring the interaction effects between characteristics and macroeconomic conditions could shed light on the underlying economic mechanisms driving these return patterns.

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